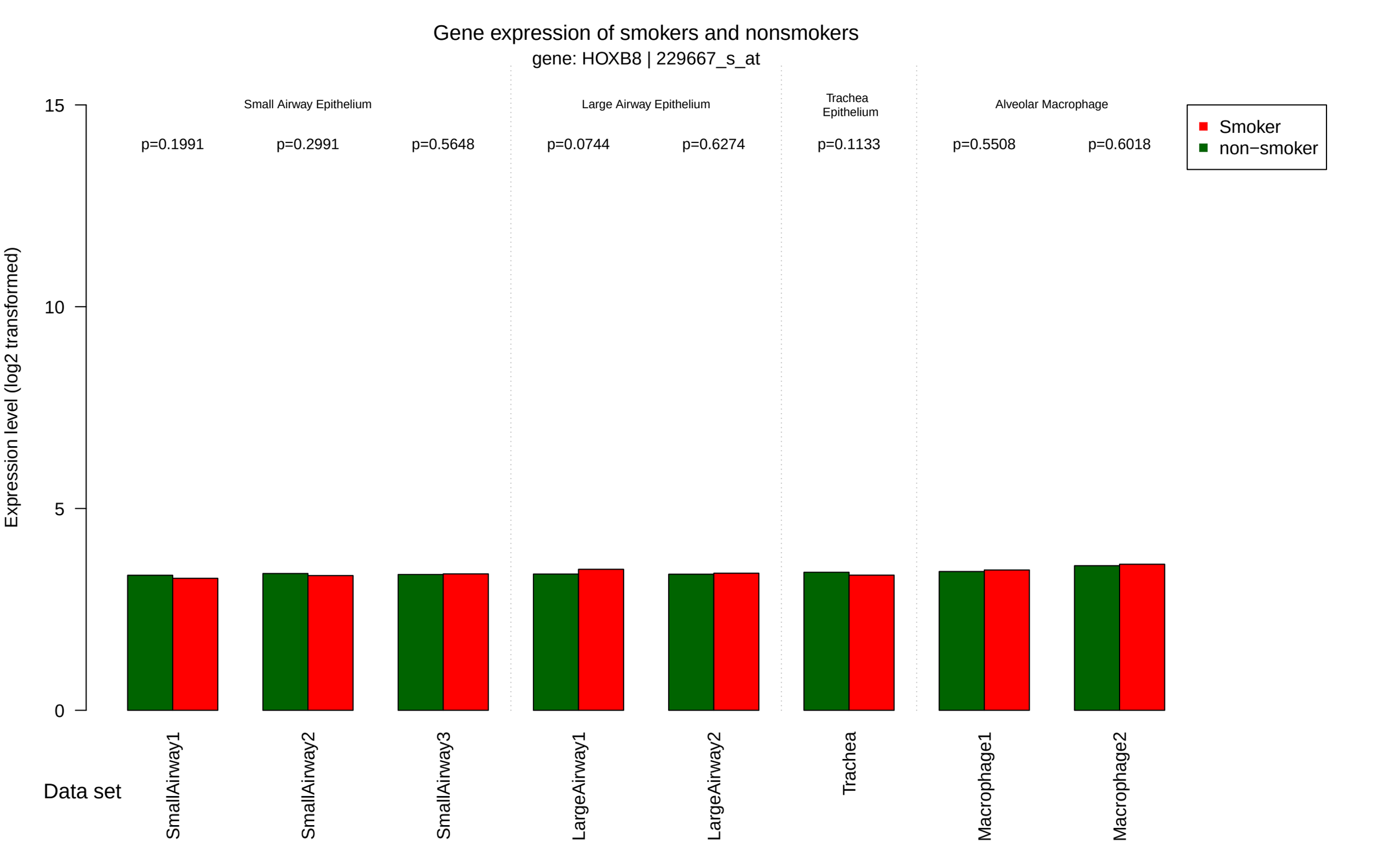
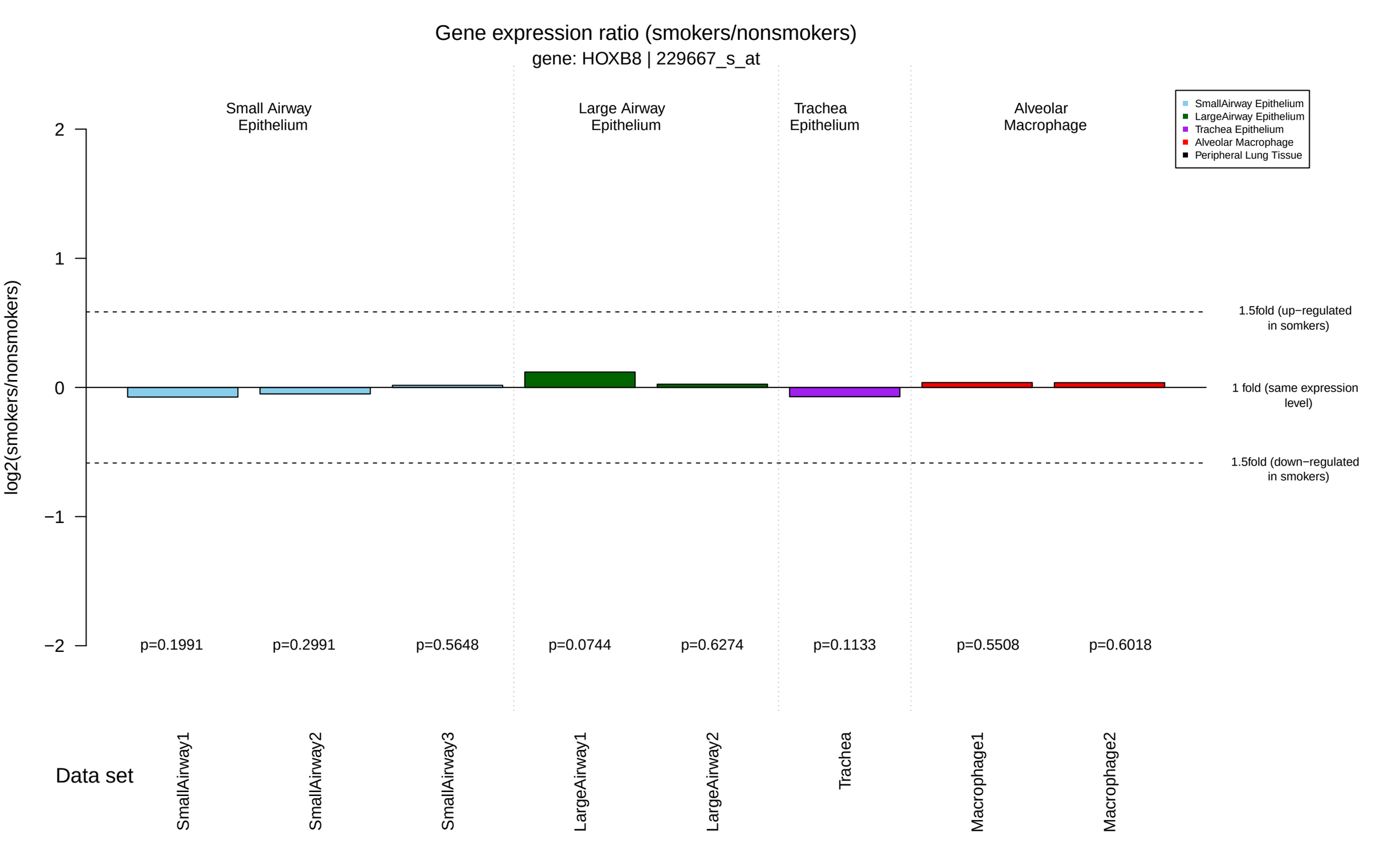
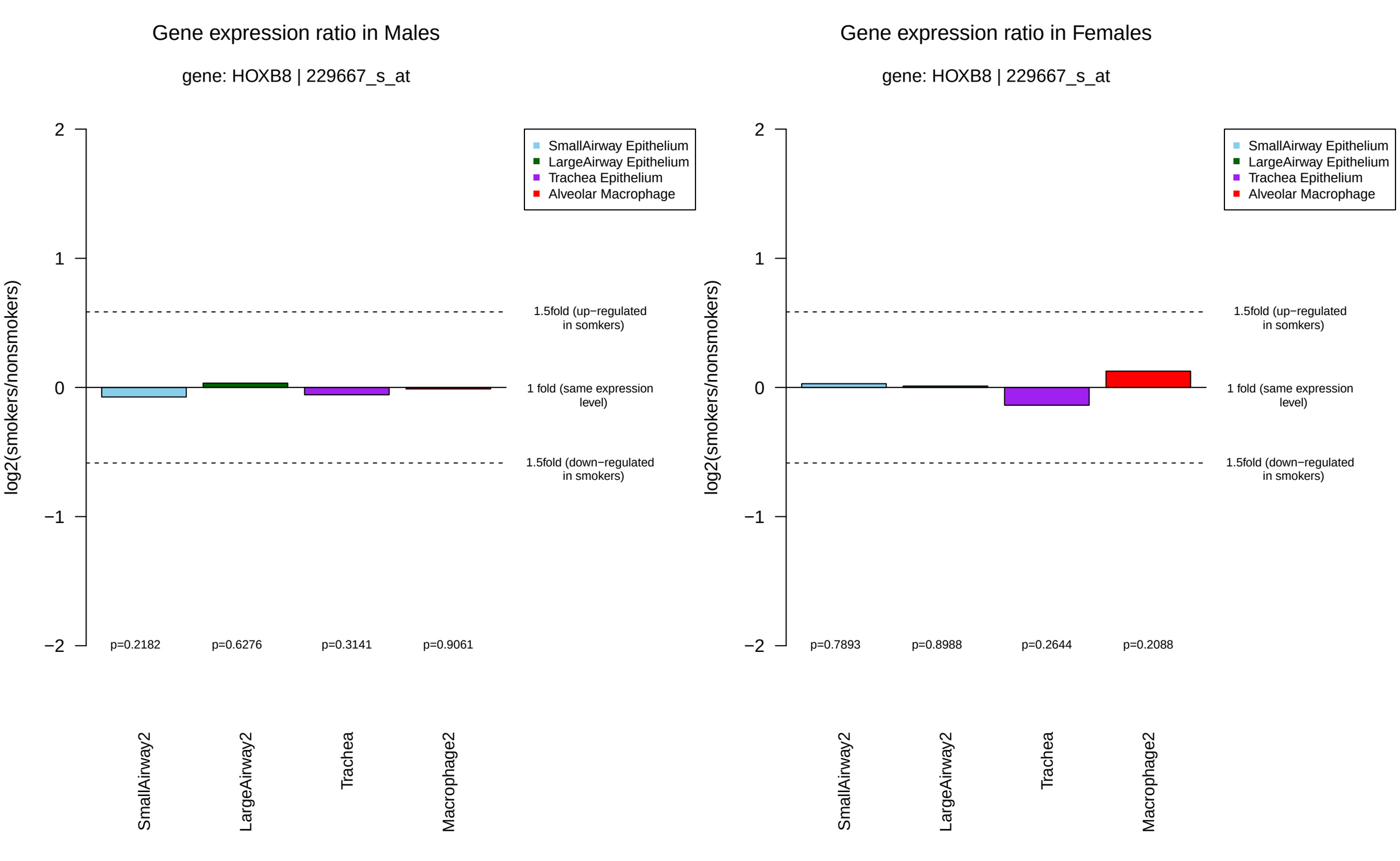
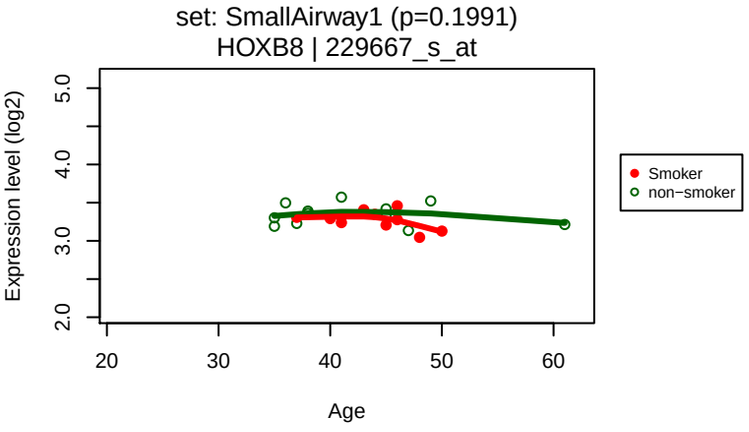
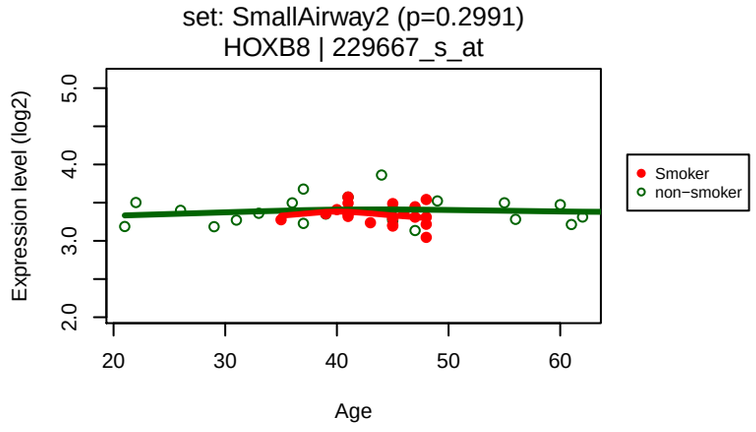
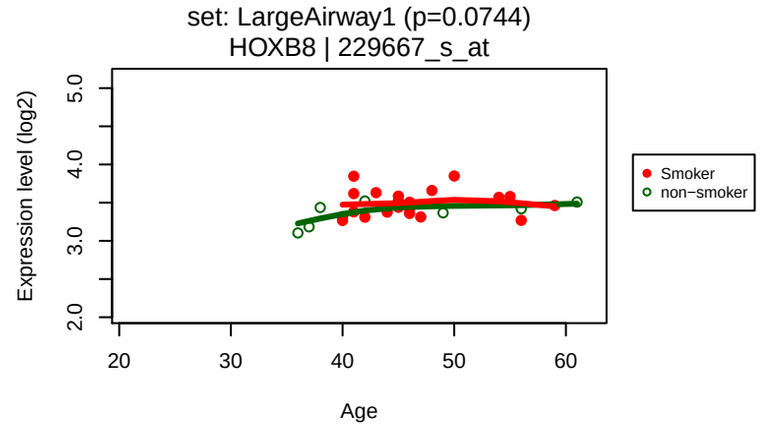

Gene expression of smokers and nonsmokers
gene: HOXB8 | 229667_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1991
p=0.2991
p=0.5648
p=0.0744
p=0.6274
p=0.1133
p=0.5508
p=0.6018
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

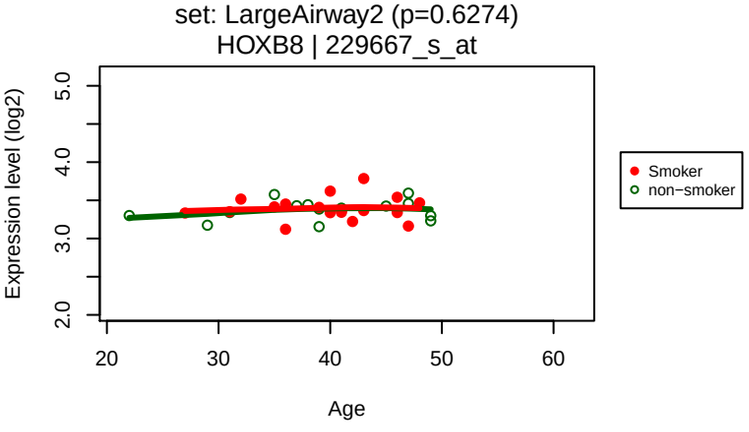

Gene expression ratio (smokers/nonsmokers)
gene: HOXB8 | 229667_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1991
p=0.2991
p=0.5648
p=0.0744
p=0.6274
p=0.1133
p=0.5508
p=0.6018
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

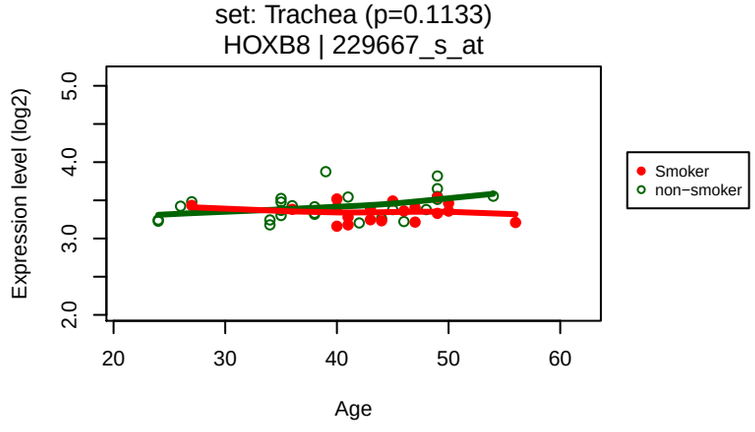

Gene expression ratio in Males
gene: HOXB8 | 229667_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2182
p=0.6276
p=0.3141
p=0.9061
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: HOXB8 | 229667_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7893
p=0.8988
p=0.2644
p=0.2088
SmallAirway2
LargeAirway2
Trachea
Macrophage2

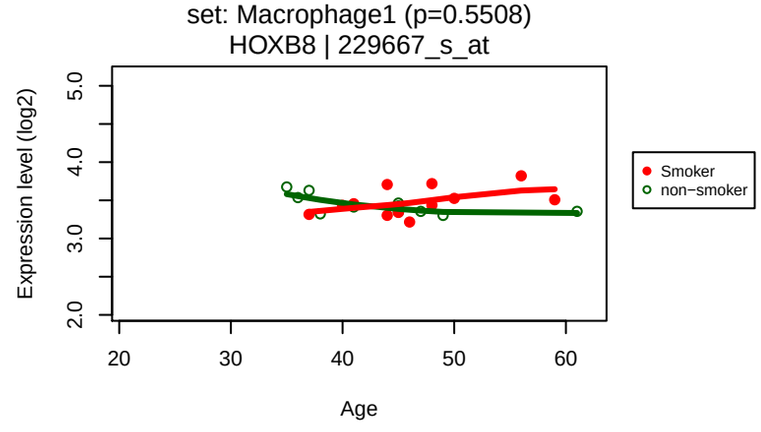

set: SmallAirway1 (p=0.1991)
HOXB8 | 229667_s_at
Expression level (log2)
Age
Smoker
non−smoker

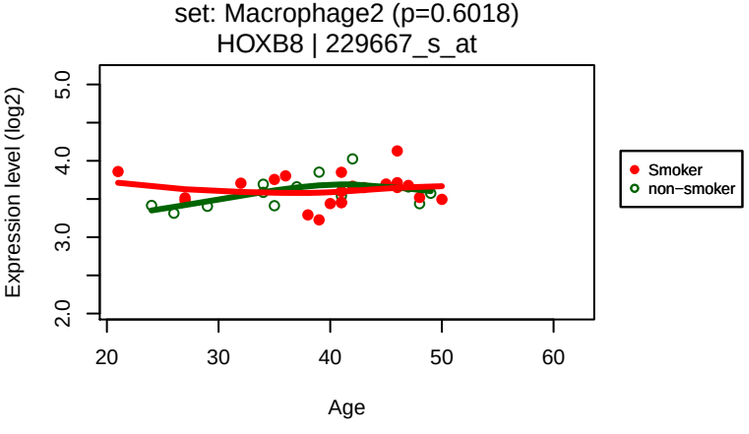

set: SmallAirway2 (p=0.2991)
HOXB8 | 229667_s_at
Expression level (log2)
Age
Smoker
non−smoker

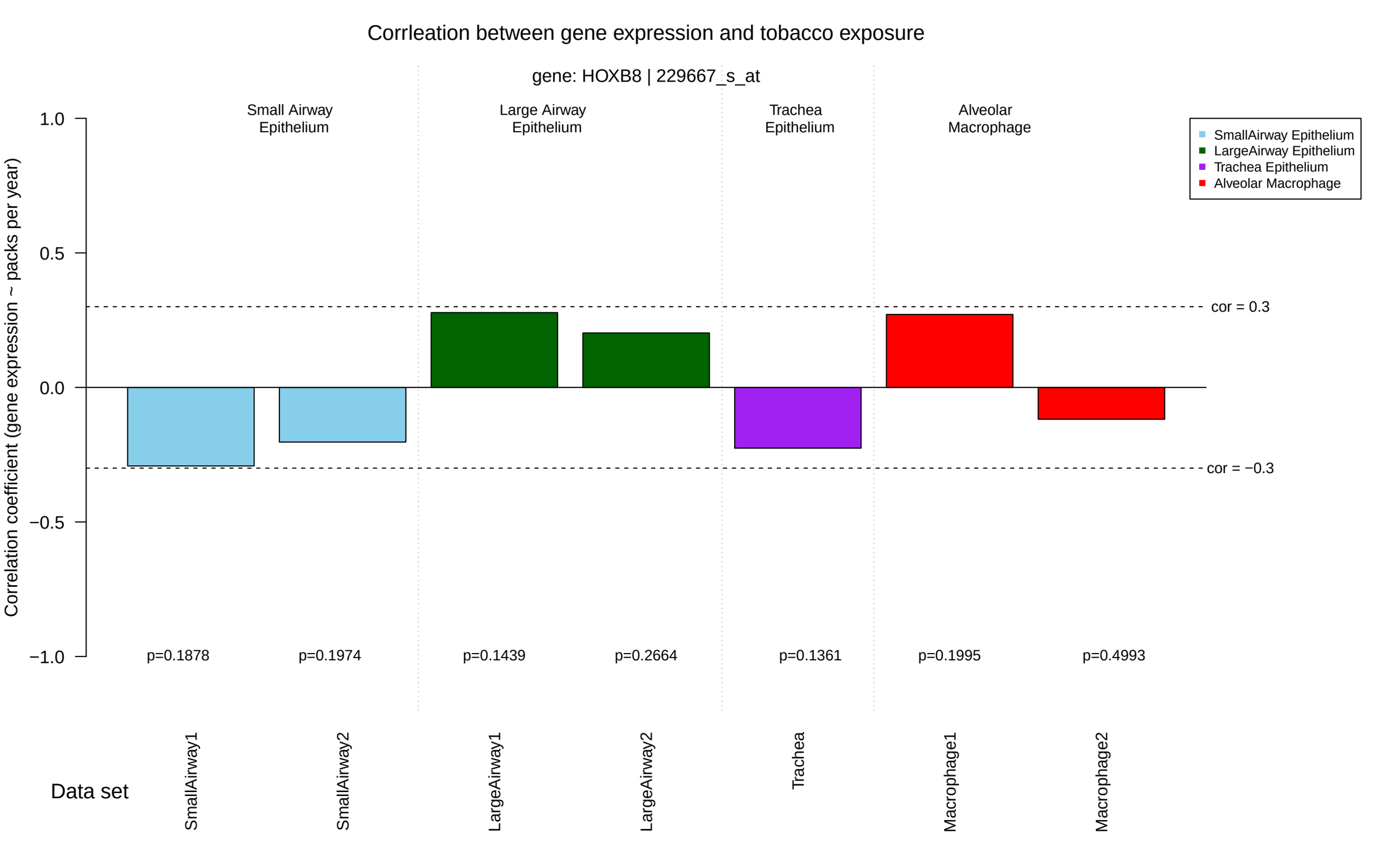

set: LargeAirway1 (p=0.0744)
HOXB8 | 229667_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.6274)
HOXB8 | 229667_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1133)
HOXB8 | 229667_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.5508)
HOXB8 | 229667_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.6018)
HOXB8 | 229667_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: HOXB8 | 229667_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1878
p=0.1974
p=0.1439
p=0.2664
p=0.1361
p=0.1995
p=0.4993
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2